

Experiment No. 6

Experiment Name

Implementation and Comparative Analysis of Image Segmentation Techniques using Python and OpenCV

Aim

To implement and evaluate various image segmentation techniques for partitioning digital images into meaningful regions, thereby facilitating object localization, scene understanding, and subsequent computer vision tasks.

Purpose

Image segmentation is one of the most important stages in computer vision, where an image is divided into multiple meaningful regions or objects based on pixel intensity, color, texture, or boundaries. Effective segmentation simplifies image analysis by isolating the regions of interest (ROI), making tasks such as object detection, medical diagnosis, autonomous navigation, and industrial inspection more accurate and efficient.

In this experiment, students will implement multiple image segmentation techniques, including Thresholding, Otsu's Thresholding, Adaptive Thresholding, Watershed Segmentation, and K-Means Clustering, and compare their performance on real-world images.

Prerequisites

- Basic knowledge of Digital Image Processing
- Familiarity with Python programming
- Python libraries: OpenCV (cv2), NumPy, Matplotlib, and Scikit-learn
- Understanding of grayscale images, image histograms, and clustering concepts
- Jupyter Notebook, Google Colab, or Visual Studio Code

Steps

1. Import the required Python libraries and load a real-world image such as a medical image, satellite image, or natural scene.
2. Convert the image into grayscale and perform basic preprocessing to reduce noise using Gaussian Blur.
3. Apply Global Thresholding to separate foreground objects from the background and observe the segmentation results.

4. Implement Otsu's Thresholding to automatically determine the optimal threshold value and compare it with global thresholding.
5. Apply Adaptive Thresholding to segment images with uneven illumination conditions.
6. Implement the Watershed Segmentation algorithm to separate overlapping or touching objects within the image.
7. Apply K-Means Clustering for color-based image segmentation by grouping pixels into distinct clusters.
8. Compare the segmentation outputs obtained from different techniques based on object boundaries, accuracy, and computational efficiency.
9. Visualize and analyze the segmented images to identify the most suitable technique for the selected dataset.
10. Summarize the observations and discuss the role of image segmentation in advanced computer vision applications.

Questions

1. What is image segmentation? Why is it considered a fundamental step in computer vision?
2. Differentiate between Image Segmentation and Image Classification.
3. Explain the working principle of Global Thresholding, Otsu's Thresholding, and Adaptive Thresholding.
4. What is the Watershed Algorithm? Why is it useful for separating overlapping objects?
5. How is K-Means Clustering applied to image segmentation?
6. Compare threshold-based segmentation and clustering-based segmentation techniques.
7. What challenges are encountered while segmenting images with complex backgrounds or varying illumination?
8. Mention five real-world applications where image segmentation plays a critical role.
9. How does image segmentation improve the performance of object detection and image recognition systems?
10. Compare traditional image segmentation techniques with deep learning-based segmentation methods such as U-Net and Mask R-CNN.